Eduardo Gaitán Escalante

Data Scientist & AI Engineer · Economist · Mexico City, Mexico (UTC-6)

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English: fluent · Spanish: native · Remote · Updated November 2026

SUMMARY

Economist (UNAM, honors) with **5+ years building data pipelines and models in Python/R** and 1+ year shipping LLM systems to production. Build Claude API agents that automate document processing and incident analysis, with structured outputs, cost controls, local-model fallback and human approvals. Combine SQL analytics, machine learning, econometrics and stakeholder reporting with end-to-end delivery, including financial analytics and a Smart CFO SaaS MVP.

CORE SKILLS

AI agent workflows: Anthropic Claude API, structured outputs, prompt design, receipt extraction, incident analysis, per-call cost control, Ollama/Qwen3-VL fallback, human-approval gates; coding-agent orchestration with tests and draft pull requests.

Analytics foundations: SQL, PostgreSQL, BigQuery, Python, R, pandas, Parquet, ETL; Kedro pipelines, MLflow experiment tracking, Google Cloud Run and Cloud Build, Docker, Git. Spark/PySpark and Databricks: certifications and one experiment.

Statistics & ML: controlled experiments, A/B test design, hypothesis testing, econometrics, out-of-sample validation, spatial-block cross-validation, forecasting (Prophet, SARIMA, XGBoost, Croston), Isolation Forest, K-Means, Monte Carlo, Sobol sensitivity.

Dashboards & reporting: Power BI, Streamlit, Plotly, Jupyter, Excel, reproducible analytical reports and investor decks; cross-functional delivery with Economics and Finance teams.

PROFESSIONAL EXPERIENCE

Independent Data Scientist & AI Engineer (freelance / contract) — Jan 2025 - Present

- Ship production agents directly on the **Claude API**: a receipt-reading accounting bot with structured outputs at approximately **US$0.02 per receipt** and a local **Qwen3-VL fallback**; a server-incident analyst whose actions require **human approval**.
- Build an agent control plane that runs coding agents, executes their tests and opens draft pull requests. Use explicit output structure, per-call cost controls and review gates to constrain production LLM workflows.
- With **Smart CFO**: develop per-SKU/branch forecasting pipelines using Kedro, MLflow, BigQuery and Cloud Run; cost/revenue analytics and opportunity dashboards. Contribute to a **SaaS MVP for business valuation and financial projections**.
- Deliver demand, pricing and capital studies as reproducible Python reports and investor decks: spatial econometrics, forecasting, price elasticity and Monte Carlo/Sobol analysis.

Naturgy | Data Science Specialist — Jan 2025 - Jul 2025

- Led a cross-functional team to deploy **Isolation Forest anomaly detection** end to end: ETL, orchestration, tuning and performance analysis in H2O and R.
- Partnered with Economics and Finance to automate **7 manual processes**, reducing processing time from **5 hours per task to under 1 minute**.
- Modernized K-Means segmentation (+15% accuracy, -10% training time); developed a hybrid XGBoost-plus-seasonality model for annual natural-gas consumption.

CONEVAL | Head of the Small Area Estimation Methodologies Department — Jan 2024 - Jan 2025

- Led a Docker/Kedro/PostgreSQL data-pipeline project; used SQL for dataset querying and manipulation, and contributed to poverty-analysis publications.
- Fitted supervised and unsupervised models: Random Forest, boosting, regression/logistic regression, SVM, neural networks and hybrid time-series models.

PROFESSIONAL EXPERIENCE (CONTINUED)

JGCDMX | Environment, Tourism & Economic Development Coordinator Oct - Nov 2023

- Automated ETL in Python, reducing project-creation time by 20%; built interactive geospatial dashboards in GeoPandas, Leaflet and Folium for decisions and route planning.
- Deployed K-Means and spatial-lag models for heatmaps, pattern analysis and reporting.

UNAM, Program for Development Studies | Python & R Developer Jan 2020 - Sep 2023

- Led a night-time-light geospatial data pipeline with Parquet for analysis, storage and reporting; conducted socioeconomic geospatial estimations.
- Built Docker development environments integrating PostgreSQL, Python and R.

SELECTED PROJECTS

Distributor forecasting pipeline | SmartCFO AI Sep 2025 - May 2026

- Kedro pipeline over approximately **4,940 branch × manufacturer series**, a **52-week horizon** and 4-week validation; Prophet, SARIMA, XGBoost and Croston/SBC. MLflow tracking, BigQuery and Cloud Build; **93× speed-up** through vectorization and a documented fix for a flat-forecast regression.

Market, demand & capital study | confidential restaurant client Sep - Oct 2026

- Validated a Huff gravity demand model out of sample: **23% median absolute error versus 74% for a naive baseline**. Used OLS/SAR/SEM with spatial-block CV, competition clustering, ETS/SARIMA forecasts and 20,000-run Monte Carlo with Sobol sensitivity.

Housing price model & decision deck | Monterrey Aug 2026

- Scraped and cleaned **936 houses across 12 municipalities**. Hedonic model fitted on 791 observations: R^2 0.85 and 19% median out-of-sample error with 5-fold CV; interpretable coefficients and a 17-slide decision deck.

Production POS platform & receipt automation | bearandchimps.com Jul - Oct 2026

- Fastify, SQLite and React/TypeScript system with order intake, kitchen screens, payments and a Claude/Qwen3-VL receipt reader. **1,300+ automated tests on the platform**; scripted deployments with verification and rollback.

EXPERIMENTATION & RELIABILITY

Controlled experiments: independently designed and ran controlled experiments; A/B test design and hypothesis testing. Designed an alternating-week free versus paid delivery protocol with fixed menu/promotions and channel-level measurement.

LLM reliability controls: structured extraction, a local-model fallback, cost tracking and approval before incident-response actions. Statistical/ML model validation includes held-out forecast validation, spatial-block CV and regression diagnosis.

EDUCATION & CERTIFICATIONS

Universidad Nacional Autónoma de México | B.A. in Economics, honors Oct 2024

- GPA 9.4/10. Coursework: Econometrics, Statistics, Time Series, Mathematics, Linear Algebra and Game Theory.
- Certifications: Machine Learning with Apache Spark & PySpark; Databricks Fundamentals; Geographic Data Science with Python (University of Chicago). Databricks project: one air-quality Gradient Boosting/MLflow experiment (Dec 2024).